Who Is Really on the Frontier? Confidence Sets for Cost-Accuracy Pareto Leaderboards

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Abstract

A leaderboard that reports cost beside accuracy is read through its Pareto frontier: the systems that no cheaper system beats. Every published frontier is drawn from point estimates of accuracy on a sample of tasks, so the frontier is itself an estimate, and a reader cannot tell a system that is on it from one that would be on it under another draw of tasks. We give a set of tools for reading such boards. A simultaneous confidence band for the frontier curve, the best accuracy available at each cost, and a confidence set for its members, which contains the whole true frontier with probability at least $1 - \alpha$; the band is proved for fixed and per-task costs alike, the set for fixed costs and for per-task costs without ties, and both are checked by simulation, against the obvious alternatives, and on held-out halves of real boards. An undominated frequency says how often each system would be on the frontier under resampling. The cheapest plausible best is the cheapest system that cannot be told apart from the most accurate one. Two certificates for a new frontier entrant come with chain-wise error control: a record claim against the strongest cheaper system, which reduces to the exact selective test of our companion paper, and a certificate of Pareto membership from an intersection-union test conditional on the whole frontier event, in a normal-approximation form and an exact randomization form; we prove that no certificate built on the entrant's strongest rival alone can hold its level, and show by simulation what each test controls and what it costs in power. The reported accuracy of a frontier member is inflated by the selection that put it there, and the winner's-curse corrections of that paper apply with the pool of cheaper systems in place of the board. In simulation the frontier set contains the whole true frontier 0.93 to 1.00 of the time, while the plain frontier wrongly includes 5 to 28% of its members and misses 4 to 39% of the true frontier. We audit the Holistic Agent Leaderboard: 11 benchmarks and 356 agent runs, with per-task outcomes and per-task costs rebuilt from the published call logs under one price table and with each model call counted once, which HAL's own totals do not do. Counting and pricing consistently changes HAL's published frontiers on 8 of 11 benchmarks, moving 28 of the 56 systems on them. With the uncertainty added, 64 systems (18%) sit on the observed frontiers, but 280 (79%) cannot be excluded from them; on 2 benchmarks no system can be excluded at all, and on 7 of the 11 the cheapest plausible best is a cheaper system than the top one. Walking the boards in submission order, 69 systems landed on a frontier on arrival and 6 of them can be certified with the chain-wise guarantee. The methods are released in the open-source package `honestboard`.

1 Introduction

Leaderboards for AI agents increasingly report what a run costs beside how often it succeeds. The Holistic Agent Leaderboard (HAL) [14] does this for every benchmark it hosts, and it draws the cost-accuracy picture the way an economist would: a system is worth looking at only if no cheaper system does at least as well. The systems that pass this test form the Pareto frontier, and HAL's own write-up stresses how few systems sit on it. Readers, buyers and the teams that build agents act on that picture. A system on the frontier is the one to recommend at its price. A system below it is the one to drop.

The picture is drawn from point estimates. Each system's accuracy is its success rate on a sample of tasks: 33 on AssistantBench, 45 on CORE-Bench hard, 50 on tau-bench airline and on the SWE-bench Verified mini split. With samples that small the rate carries a standard error of several points, and the frontier, which is a function of those rates, moves when the rates move. A system that sits just below the frontier might sit on it under another draw of tasks, and a system that sits on it might not. HAL reports no uncertainty for its frontiers, and we know of no leaderboard that does.

This paper supplies that uncertainty. It answers five questions.

1. How high can the frontier curve, the best accuracy available at each cost, really be, and how low; and which systems cannot be excluded from it at a stated confidence level?
2. How wrong is the plain frontier, when the truth is known?
3. What is the cheapest system a buyer cannot tell apart from the best?
4. When a new system arrives and lands on the frontier, how can a maintainer certify that it beats the strongest cheaper system, with a guarantee that survives the fact that it was tested because it landed there?
5. By how much does being on the frontier inflate a system's reported accuracy, and what is the corrected number?

What is new. The frontier itself gets a confidence band (Section 3.2): a simultaneous band for the curve that gives, at each cost, the best accuracy any system at that cost or less can truly have. It is the object every cost-aware leaderboard draws, and none has drawn it with its uncertainty. The confidence set for the frontier (Section 3.3) is a multiple-comparisons-with-the-best construction [12] with each system's rival set restricted to the systems that cost no more than it does. It is a confidence set in the full sense: it contains the whole true frontier with probability at least $1 - \alpha$, proved for fixed costs (Proposition 2) and checked by simulation, against six alternative constructions, and on held-out halves of the real boards, when costs are measured per task and resampled with the outcomes. The certification rule (Section 3.7) rests on an observation that removes what looked like a new selective-inference problem: a new system is on the frontier exactly when it beats the most accurate system that is cheaper than it, so certifying a frontier entrant is the record test of our companion paper [5] against that one holder, and the guarantees proved there apply (Proposition 4, Corollary 1). We say exactly what that certificate establishes, a record claim within the cheaper pool, prove that no certificate built on the entrant and its strongest rival alone can hold its level (Proposition 7), and give certificates that do establish Pareto membership: an intersection-union test of polyhedral selective p-values conditional on the whole frontier event (Proposition 5) and an exact randomization version of it that conditions on that event by keeping only the outcome swaps that leave the entrant on the frontier (Proposition 6), with a simulation

of what each controls and what each costs in power (Table 1). An exact finite-sample frontier set (Section 3.3) and a resolution number per board, the task count at which a board would start to resolve its own frontier (Section 3.3), complete the set of tools. The winner's-curse corrections of the companion paper carry over to frontier members with the cheaper pool in place of the board (Section 3.8). The cheapest plausible best (Section 3.5) turns the confidence set of the best into the number a buyer asks for. The audit (Section 6) rebuilds per-task costs for 356 HAL runs from their call logs, under one price table and with each model call counted once, which is itself a contribution: HAL's reported totals embed the prices in force when each run was made and double count nested trace spans on about half the runs, so two runs are not comparable without repricing and recounting.

Related work. Uncertainty in leaderboard rankings has been treated through confidence sets for ranks [19–21], confidence sets of the best [11, 12], the winner's curse of a selected system [1, 4], and standard errors for language-model evaluations [8, 17, 18]. The companion paper [5] gives selective exact tests for leaderboard records and online error control over a chain of them, building on selective inference [10, 15] and online family-wise error control [22]. Cost has entered leaderboard design through HAL [14], the argument that agent evaluations should report it [13], and the cost-of-pass framework, which prices a correct answer at the frontier of models and reports how that frontier moves [9], all from point estimates. Identifying a Pareto set from noisy observations has been studied where the analyst chooses what to sample next: Pareto-set identification with (ϵ, δ) guarantees from bandit feedback [3], active learning of the Pareto front [24], and multi-objective ranking and selection in simulation [2]. A leaderboard is the fixed-sample version of that problem, with every system scored on the same tasks, with paired outcomes, and with systems arriving over time and being announced as frontier members by the rule that selected them; the confidence set, the selective certification and the audit here are built for that setting, and we know of no earlier treatment of a published cost-accuracy frontier as an estimated object with a confidence statement attached.

Findings. HAL's cost totals double count nested trace spans on 202 of 332 runs (141 of them exactly doubled) and carry the prices of the month each run was made; counted once and priced under one table, the point-estimate frontier changes on 8 of 11 benchmarks and 28 of the 56 systems on HAL's published frontiers change status. On the corrected boards (356 agent runs), 64 systems (18%) sit on the observed frontier, but 280 (79%) cannot be excluded from it at the 5% level. The share that cannot be excluded runs from 42% to 100% across benchmarks: it is smallest on USACO and it is everything on AssistantBench and SciCode, the 2 boards with few tasks and low accuracies. The best set is a single system on 2 boards. On 7 of 11 the cheapest plausible best is a cheaper system than the top one and the best set is smaller than the board, sometimes much cheaper: on tau-bench airline the top system costs \$3.14 per task at 66.0% and the cheapest plausible best costs \$0.04 at 44.0%, a gap the 50 tasks cannot resolve. Walking the boards in submission order, 69 systems landed on a frontier on arrival; 12 of them pass an unadjusted test against their strongest cheaper rival, 6 can be certified as record claims with the family-wise guarantee over the chain, and 8 as Pareto members. Excluding half of the systems that cannot now be excluded would take 122 to 7,863 tasks per board. In simulation the frontier set keeps a truly undominated system 0.987 to 1.000 of the time, while the plain frontier wrongly includes 5 to 28% of its members and misses 4 to 39% of the true frontier.

2 Setting

A board has K systems evaluated on the same T tasks. System k scores $y_{kt} \in [0, 1]$ on task t: a pass or fail on most benchmarks, a partial score on some. Its accuracy is the mean $\bar{y}_k = T^{-1} \sum_t y_{kt}$, which estimates $\mu_k = \mathbb{E}\, y_{kt}$. Its cost is either one number per system, c_k (the price of the run divided by T), or one number per task, c_{kt}, with $\bar{c}_k$ the mean; on HAL the per-task costs come from the call logs (Section 4).

System j *dominates* system i when $c_j \le c_i$ and $\bar{y}_j \ge \bar{y}_i$ with at least one of the two strict. The *observed frontier* is the set of systems that no other system dominates. Two systems equal in both cost and accuracy dominate neither the other, so both can be on the frontier; this is the only tie convention used, and it is applied throughout. The *true frontier* is the same set with μ_k in place of $\bar{y}_k$: a system is truly undominated when no system that costs no more than it has a higher true accuracy.

The randomness we model is the draw of tasks. The T task vectors $(y_{1t}, \ldots, y_{Kt})$ are treated as exchangeable draws from the benchmark's task population, dependent across systems on the same task (hard tasks are hard for everyone) and independent across tasks. The price table, dollars per token for each model, is a fixed list; a system's cost is the priced token count of its run, which varies with the tasks drawn, so costs are random where prices are not. Three things are not modelled, and Section 7 returns to them: tuning of a system on the public tasks, run-to-run randomness of a system on the same task, and changes in prices over time.

3 Methods

3.1 The observed frontier and why it is not enough

The observed frontier is what every cost-aware leaderboard draws. It is a function of K estimated accuracies, each with a standard error of order $T^{-1/2}$, and it is a discontinuous function: moving one accuracy by a hair can add or remove a system. Section 5 measures how often it is wrong when the truth is known. The short version is that with 300 simulated boards of 20 systems on 50 tasks, 21% of the systems on the observed frontier are truly dominated and 29% of the true frontier is missed.

3.2 A confidence band for the frontier curve

What a cost-aware leaderboard draws is a curve: at each cost x, the best accuracy available from any system whose cost is at most x,

$$A(x) = \max_{j:\, \gamma_j \le x} \mu_j,$$

with γ_j the true mean cost of system j: a staircase that steps up at the cost of each frontier member. The plain frontier is this curve drawn with the sample means $\bar{y}_j$ and $\bar{c}_j$ in place of μ_j and γ_j. A simultaneous band for A follows from a simultaneous band for all $2K$ means at once. Let se_j and sg_j be the standard errors of $\bar{y}_j$ and $\bar{c}_j$ (zero for a cost that is a fixed number), and let c_{band} be the $(1-\alpha)$ quantile, over task resamples that carry outcomes and costs together, of the largest of $|\bar{y}_j^{(b)} - \bar{y}_j|/\mathrm{se}_j$ and $|\bar{c}_j^{(b)} - \bar{c}_j|/\mathrm{sg}_j$ over all systems. Each system then has an accuracy interval $[\bar{y}_j \mp c_{\mathrm{band}}\mathrm{se}_j]$ and a cost interval $[\bar{c}_j \mp c_{\mathrm{band}}\mathrm{sg}_j]$, and

$$L(x) = \max_{j:\ \bar{c}_j + c_{\mathrm{band}}\mathrm{sg}_j \le x} (\bar{y}_j - c_{\mathrm{band}}\mathrm{se}_j), \qquad U(x) = \max_{j:\ \bar{c}_j - c_{\mathrm{band}}\mathrm{sg}_j \le x} (\bar{y}_j + c_{\mathrm{band}}\mathrm{se}_j).$$

A system enters the lower edge only when its cost is surely within the budget x and the upper edge when it possibly is, so neither edge relies on having the cost order right.

Proposition 1 (band)**.** *Let the task vectors of outcomes and costs be i.i.d. with finite, positive variances, and let K be fixed as $T \to \infty$. Then*

$$\mathbb{P}(L(x) \le A(x) \le U(x) \text{ for every } x) \;\ge\; 1 - \alpha - o(1),$$

with costs fixed or measured per task, and with no condition on ties among the true mean costs.

Proof. Let E be the event that every μ_j lies in its accuracy interval and every γ_j in its cost interval. If $\bar{c}_j + c_{\text{band}}\text{sg}_j \le x$ then on E $\gamma_j \le x$, so $\bar{y}_j - c_{\text{band}}\text{se}_j \le \mu_j \le A(x)$; hence $L(x) \le A(x)$. If $\gamma_j \le x$ then on E $\bar{c}_j - c_{\text{band}}\text{sg}_j \le \gamma_j \le x$, so the maximizer j of μ_j over $\{\gamma_j \le x\}$ enters $U(x)$ with $\bar{y}_j + c_{\text{band}}\text{se}_j \ge \mu_j = A(x)$; hence $U(x) \ge A(x)$. Both hold for every x on E. By the multivariate central limit theorem and the consistency of the bootstrap for the mean vector [7], with the standard errors converging in probability by Slutsky's theorem, the bootstrap law of the maximal studentised deviation converges to that of the maximum of the absolute values of a non-degenerate Gaussian vector, whose distribution function is continuous, so $\mathbb{P}(E) \ge 1 - \alpha - o(1)$. □

Nothing in the proof uses the cost order, so the band holds where the frontier set of Section 3.3 needs the order to be recoverable. The price is width: a system whose cost is uncertain is allowed into the upper edge long before it is into the lower one. The band is a statement about the curve, the set a statement about its members; the two use different statistics and neither implies the other. The band is what to draw on a leaderboard page, Figure 1; its width on HAL, as a median over the evaluated costs, runs from 6 points (ColBench frontend) to 67 points (CORE-Bench hard), and in simulation, with per-task costs resampled and true mean costs closely spaced, it covers the true curve 0.96 to 0.99 of the time (Table 4). A first version of the band that trusted the sample cost order covered the true curve only 18 to 62% of the time on those boards, which is why the construction above lets a system into an edge only on the strength of its cost interval.

3.3 A confidence set for the frontier

Call j a *rival* of i when $c_j \le c_i$ and $j \ne i$: a rival is any system that could dominate i on cost. For each pair let $\widehat{\text{se}}_{ij}$ be the paired standard error of $\bar{y}_j - \bar{y}_i$, computed from the task-level covariance of the two outcome series, and let

$$G_{ij} = \frac{\bar{y}_j - \bar{y}_i}{\widehat{\text{se}}_{ij}}$$

be how far j sits above i in standard-error units. System i is on the observed frontier exactly when $G_{ij} < 0$ for every rival (with the tie conventions). The confidence set keeps i unless some rival beats it by more than a simultaneous margin:

$$\widehat{F}_\alpha = \Big\{ i : \max_{j \text{ rival of } i} G_{ij} \le c_\alpha \Big\}.$$

The margin is a bootstrap quantile. Resample the T tasks with replacement B times; in resample b write $\bar{y}_k^{(b)}$ for the accuracies and, when costs are per task, $\bar{c}_k^{(b)}$ for the costs, so that the rival sets are resampled too. Let

$$M_i^{(b)} = \max_{j \text{ rival of } i \text{ in resample } b} \frac{(\bar{y}_j^{(b)} - \bar{y}_i^{(b)}) - (\bar{y}_j - \bar{y}_i)}{\widehat{\text{se}}_{ij}},$$

the largest centred, studentised deviation of any rival from i, and let $M^{(b)} = \max_i M_i^{(b)}$ be the largest such deviation on the board. The margin is the $(1-\alpha)$ quantile of $M^{(b)}$ over resamples, $c_\alpha = \max(0, q_{1-\alpha}(M))$. This is Hsu's multiple-comparisons-with-the-best construction [12], as implemented for the confidence set of the best in Banahene and Opoku [5], with the rival set of each system cut down to the systems that cost no more and the margin taken jointly over systems. Two properties are immediate: $\widehat{F}_\alpha$ always contains the observed frontier, because a frontier member has every $G_{ij} < 0 \le c_\alpha$; and it shrinks toward the observed frontier as T grows, because the margin is in standard-error units.

Proposition 2 (the set contains the true frontier)**.** *Fix the costs. Let the task vectors be independent and identically distributed with finite variances, with the population variance of $y_{jt} - y_{it}$ positive for every pair compared, and let K be fixed as $T \to \infty$. Let F be the set of systems that no rival beats in true accuracy, the i with $\mu_j \le \mu_i$ for every rival j; it contains the true frontier of Section 2, which also drops a system tied in true accuracy with a strictly cheaper one. Then*

$$\mathbb{P}(F \subseteq \widehat{F}_\alpha) \;\ge\; 1 - \alpha - o(1).$$

Proof. For $i \in F$ and any rival j, $\mu_j - \mu_i \le 0$, so $G_{ij} \le Z_{ij} := \{(\bar{y}_j - \bar{y}_i) - (\mu_j - \mu_i)\}/\widehat{\mathrm{se}}_{ij}$, and therefore $\max_{i \in F} \max_j G_{ij} \le \max_{i \in F} \max_j Z_{ij} \le \max_i \max_j Z_{ij} =: Z$, each maximum over the system's rivals, sets that do not change with the data when costs are fixed. By the multivariate central limit theorem the centred, scaled mean vector is asymptotically Gaussian, and the nonparametric bootstrap of the mean vector is consistent [7]. Z and M both divide by the sample standard errors $\widehat{\mathrm{se}}_{ij}$, which converge in probability to their population values, so by Slutsky's theorem and the continuous mapping theorem Z and the bootstrap M share the limit law of the maximum of a non-degenerate Gaussian vector, whose distribution function is continuous and strictly increasing; hence c_α converges to that law's $(1-\alpha)$ quantile and $\mathbb{P}(Z > c_\alpha) \to \alpha$. Hence $\mathbb{P}(F \not\subseteq \widehat{F}_\alpha) = \mathbb{P}(\max_{i \in F} \max_j G_{ij} > c_\alpha) \le \mathbb{P}(Z > c_\alpha) \le \alpha + o(1)$. □

Remark 1. *This is a confidence set for the frontier as a set: with probability about $1-\alpha$ every truly undominated system is kept at once. Two smaller sets with weaker, per-system guarantees come from the same resamples: the largest per-system quantile, $\max_i q_{1-\alpha}(M_i)$, as the margin, and each system's own quantile $q_{1-\alpha}(M_i)$; each keeps a given truly undominated system with probability about $1-\alpha$. Section 5 reports all three against the alternatives a practitioner might reach for instead, and on HAL the three sets hold 280, 259 and 248 systems. The joint set is the one the rest of the paper calls the frontier set.*

Proposition 3 (per-task costs)**.** *Let costs be measured per task, with the task vectors of outcomes and costs i.i.d. with finite variances, and let the true mean costs $\gamma_1, \ldots, \gamma_K$ be distinct. Let F be defined from the true accuracies and true mean costs. Then the frontier set built with per-task costs resampled jointly with the outcomes satisfies $\mathbb{P}(F \subseteq \widehat{F}_\alpha) \ge 1 - \alpha - o(1)$. (The band of Proposition 1 needs no such condition.)*

Proof. With distinct true mean costs, the sample mean costs recover the true cost order with probability tending to one, and so do the resampled mean costs within every resample except a vanishing share, since each is $\sqrt{T}$-consistent for its target. On that event every rival set, real and resampled, equals the true rival set, and the statistics coincide with those of the fixed-cost construction with the true rival sets; Propositions 2 and 1 then apply. Ties in true mean cost are the case the proof excludes: a rival set that stays random in the limit. □

Remark 2. *On HAL the per-task and fixed-cost versions give identical sets on 10 of the 10 boards with per-task costs (Section 6), because cost differences between neighbouring systems are mostly many times their standard error, which is the no-ties condition in practice.*

An exact set for any number of tasks. Proposition 2 is asymptotic. A finite-sample companion keeps i unless some rival j beats it in a one-sided exact paired test at level α/m_i, with m_i the number of i's rivals: the sign test on the discordant tasks for pass/fail outcomes, a sign-flip test for rates. A truly undominated system has no rival that is truly better, so under the task-wise null each rival's test rejects with probability at most α/m_i and the system is kept with probability at least $1-\alpha$, for any T, by Bonferroni over its rivals; dividing by the number of rival pairs on the whole board instead gives a joint version, which contains the whole true frontier with probability at least $1-\alpha$ for any T. The price is power: the exact sets are the bootstrap set's conservative cousins, and on HAL they hold 269 (per system) and 291 (joint) systems against 280 (Table 9).

What it would take to resolve a board. A system that is in the set but off the plain frontier is one whose strongest cheaper rival leads it by G standard errors with $G \le c_\alpha$. The lead in standard-error units grows like $\sqrt{T}$, so with T' tasks it would be $G\sqrt{T'/T}$, and the system leaves the set once $T' > T(c_\alpha/G)^2$. Two summaries of this per board are reported in Section 6: the task count at which half of the currently unexcluded systems would be excluded, and the task count at which a cheaper system leading by five accuracy points would be, given the board's typical paired noise. Neither is a sample-size formula for a new benchmark, since the gaps on a future board are unknown; they say how far the present boards are from resolving the frontiers they draw.

3.4 Undominated frequency

The share of task resamples in which system i is on the observed frontier, with costs resampled when they are per task. It is a resampling frequency, reported as a description of how stable each system's frontier membership is, not as a probability that the system is truly undominated. In simulation it averages 0.62 over truly undominated systems and 0.10 over truly dominated ones with 20 systems on 50 tasks (Table 4).

3.5 The cheapest plausible best

The confidence set of the best [5, 12] is the set of systems that cannot be excluded as the most accurate, with no reference to cost: the construction of Section 3.3 with every other system as a rival. The *cheapest plausible best* is its cheapest member. It answers the buyer's question, "as good as the best, as cheap as possible", with the honest reply "this is the cheapest system the data cannot tell apart from the best at level α". Its coverage follows from the set's: the true best is in the set with probability at least $1-\alpha$, so with that probability the cheapest plausible best costs no more than the true best.

3.6 Ingredients from the companion paper

Five pieces of Banahene and Opoku [5] are used below, each stated here in the form needed.

The selective exact p-value. For a holder a and a challenger b on the same T tasks with pass/fail outcomes, let n_b be the number of tasks b passes and a fails, n_a the reverse, and $n = n_a + n_b$. A leaderboard tests b only because $\bar{y}_b > \bar{y}_a$, that is $n_b > n_a$. The selective exact p-value is the tail of the sign test conditional on that event, $p_{\text{sel}} = \mathbb{P}(X \ge n_b \mid X > n/2)$ with $X \sim \text{Bin}(n, 1/2)$. For outcomes that are rates in $[0,1]$ the same conditioning is applied to a sign-flip test of the paired differences by Monte Carlo over 200,000 sign patterns.

The conditional task-wise dominance null. The null under which p_{sel} is valid: on every task, given the two systems' outcomes on the other tasks, b is no more likely to pass than a, with tasks independent. It is the natural "b is not better than a" for paired data. Proposition 1 of the companion paper shows that under this null p_{sel} is super-uniform given the record event, provided the holder was chosen by an event that is increasing in the holder's own outcomes and does not involve b.

The per-date factor and the best-first order. Submissions that share a date are ordered best first, so the challenger tested on a date is the best of that date, a second selection the record event does not cover. The p-value is multiplied by g, the number of submissions on the challenger's date, a Bonferroni factor over the date's members, which restores the guarantee.

Alpha-spending. The t-th tested step is tested at level $\alpha_t = 6\alpha/(\pi^2 t^2)$. The levels sum to α, so by the union bound the probability of certifying any null step over the whole chain is at most α, for any number of steps and any dependence, provided each step's p-value is valid given that the step occurred and the step's level was settled before its outcomes were looked at.

The online fallback. The t-th step is tested at α_t plus the level of the previous step when that step was certified [22, 23], so a run of certified steps accumulates budget. The family-wise guarantee is the same, by a disjointness argument over the runs of certified steps that feed each null step.

3.7 Certifying a frontier entrant

A live board changes when submissions arrive. A new system b lands on the frontier when it beats every system cheaper than it; the maintainer who then tests it has selected it for exactly that, and a test that ignores the selection is biased. The frontier event looks harder than the record event because it is an intersection of events against several systems. As an event, it is not.

Let $C(b) = \{j \text{ earlier} : c_j \le c_b\}$ be the cheaper-or-equal systems when b arrives, and let j^* be the most accurate member of $C(b)$, with ties broken by a rule that does not look at b. Then

$$b \text{ is on the observed frontier on arrival} \quad \Longleftrightarrow \quad \bar{y}_b > \bar{y}_{j^*},$$

because beating the most accurate cheaper system is the same as beating every cheaper system (a tie with a strictly cheaper system leaves b dominated, a tie at equal cost leaves both on the frontier, and b counts as an entrant only when the inequality is strict). The frontier event is therefore the record event of the companion paper for the single pair (holder j^*, challenger b), and the holder was chosen by an event, "j^* is the most accurate system in $C(b)$", that is increasing in j^*'s own outcomes task by task, depends otherwise only on the other cheaper systems, and does not involve b. These are the conditions of that paper's Proposition 1.

Proposition 4 (frontier entrant)**.** *Fix the costs in advance of the outcomes. Under the conditional task-wise dominance null for the pair (j^*, b) and independence across tasks, the selective exact p-value $p_{\text{sel}}(b \text{ vs } j^*)$ is super-uniform conditional on the event that b is on the observed frontier on arrival. A rejection at level α certifies that b beats the system that was the most accurate cheaper system when b arrived.*

Remark 3 (what the certificate says, and what it does not)**.** *The null tested is pairwise. If some cheaper system k truly dominates b while the observed best cheaper system j^* is a weak relative of b (the same agent with a smaller model, a lower effort setting, or a rerun, whose outcomes are*

correlated with b's), then b can beat j^* with the stated guarantee and still be truly dominated by k. The test is not of level α for the composite null "some cheaper system dominates b". In simulation with such a near-copy as the apparent rival and a truly dominating independent system in the pool, a dominated entrant is certified in 3.1 to 33.3% of the cases in which it lands on the frontier, against 1.1 to 3.3% when every cheaper system is independent of it (20000 boards per case). The certificate is therefore a record claim restricted to the cheaper pool, and it becomes a certificate of Pareto membership only under an added assumption: that whenever b is truly dominated by a cheaper system, it is task-wise dominated by the observed best cheaper system as well. Testing b against each cheaper system with the pairwise selective test and requiring every test to reject does not repair this, because each pairwise test conditions on its own event and not on the whole frontier event. Conditioning every test on the whole event does, at the price of a normal approximation; that is the next proposition.

Proposition 5 (Pareto certificate)**.** *Let the vector of accuracies be approximately normal, $\bar{y} \sim N(\theta, \Sigma)$ with Σ the task-level covariance divided by T, and let $E = \{\bar{y}_b \geq \bar{y}_k$ for every $k \in C(b)\}$ be the frontier event. For each $k \in C(b)$ let p_k be the polyhedral selective p-value of Lee et al. [15] for $H_k : \theta_b \leq \theta_k$ conditional on E, with the lattice correction of the companion paper. Then $p_{\mathrm{IU}} = \max_{k \in C(b)} p_k$ is super-uniform, conditional on E, under the composite null "some system in $C(b)$ truly dominates b", and rejecting when $p_{\mathrm{IU}} \leq \alpha$ certifies at level α that no cheaper-or-equal system is truly at least as accurate as b: that b is on the true frontier of its pool.*

Proof. Under the composite null some H_k holds. E is a polyhedron in $\bar{y}$, so by the polyhedral lemma p_k is uniform conditional on E when $\theta_b = \theta_k$ and super-uniform when $\theta_b < \theta_k$. Hence $\mathbb{P}(p_{\mathrm{IU}} \leq \alpha \mid E) \leq \mathbb{P}(p_k \leq \alpha \mid E) \leq \alpha$. An intersection-union test needs no correction for the number of rivals [6]. □

Two things sit outside the proof. The covariance is a plug-in estimate, and the lattice correction is a continuity adjustment for accuracies that live on the grid $1/T$: it relaxes each constraint by half a step and takes the tail from half a step below the observed value, one part conservative and one not. The null rows of Table 1, at most 3.9% and 4.1%, are their check. The two certificates answer different questions and have different prices. The pairwise test is exact for any T and asks whether b beat the system it displaced; the composite test is exact only under normality with a plug-in covariance and asks whether b beats every cheaper system at once, conditioning on an event that truncates the statistic more. Table 1 shows both on the three simulated worlds: with a near-copy rival the pairwise test certifies a truly dominated entrant in 3.1 to 33.3% of the cases in which it lands on the frontier, the composite test in 1.8 to 3.9%; with independent rivals the pairwise test certifies 1.1 to 3.3% and the composite test 1.9 to 4.1%; and when the entrant is truly better than every cheaper system by half a logit the pairwise test certifies it 10 to 77% of the time and the composite test 14 to 79%. The composite test gives up nothing in power for its stronger claim: the exact sign test is conservative because its statistic is discrete, which shows in its null rates (1.1 to 3.3% against the composite test's 1.9 to 4.1%), and the normal statistic has no such step. What the composite test gives up is exactness. The audit reports both.

Proposition 6 (exact Pareto certificate)**.** *Fix the costs. For each $k \in C(b)$, let the pair (b, k) satisfy the sharp task-wise null that on every task the two systems' outcomes are exchangeable, independently across tasks. Draw a random subset of tasks and swap b's and k's outcomes on it; keep the swap only if b still beats every system in $C(b)$ on the swapped accuracies; let p_k^{ex} be the share of kept swaps (the observed configuration counted once) on which b's lead over k is at least the observed lead. Then p_k^{ex} is uniform on its support conditional on the frontier event under*

World	Rival shape	Tasks	Boards on frontier	Certified at 5% pairwise	composite	exact composite
independent	independent rivals	50	7815	1.1%	2.0%	1.2%
independent	independent rivals	100	8045	1.2%	1.9%	1.0%
independent	independent rivals	150	9228	2.5%	3.4%	2.3%
independent	independent rivals	300	9626	3.3%	4.1%	3.4%
near-copy	near-copy, flip 0.2	50	7515	3.6%	1.9%	0.9%
near-copy	near-copy, flip 0.2	100	8065	4.9%	1.8%	1.1%
near-copy	near-copy, flip 0.2	150	9127	3.1%	3.3%	2.6%
near-copy	near-copy, flip 0.05	300	9554	33.3%	3.9%	0.5%
better	entrant truly better	50	16485	9.7%	14.5%	11.1%
better	entrant truly better	100	18579	23.7%	29.4%	24.8%
better	entrant truly better	150	19303	41.4%	46.5%	42.2%
better	entrant truly better	300	19924	76.8%	79.2%	77.3%

Table 1: What the two certificates control, 20000 boards per row. In the two null worlds a cheaper independent system with the entrant's true accuracy dominates it; the near-copy rows add a degraded copy of the entrant to the pool. In the last block the entrant is truly better than every cheaper system by half a logit. Rates are over the boards on which the entrant landed on the observed frontier. A valid Pareto certificate must stay at or below 5% in the null rows.

that null, super-uniform under the task-wise dominance null for (b, k)*, and* $p^{\mathrm{ex}}_{\mathrm{IU}} = \max_{k \in C(b)} p^{\mathrm{ex}}_k$ *is super-uniform conditional on the frontier event under the composite null, for any number of tasks.*

Proof. Under exchangeability within tasks, every assignment of the two outcomes on each task to b and k is equally likely given the unordered pairs and the other systems' outcomes, so every swap pattern is equally likely; the frontier event is a function of the pattern, and conditioning on it leaves the kept patterns equally likely. The p-value is then the randomization p-value of a finite uniform set, which is exactly uniform on its support with the observed configuration included. Super-uniformity under dominance follows as in Proposition 1 of the companion paper, and the intersection-union step as in Proposition 5. □

The exact certificate asks nothing of normality, a covariance estimate or a lattice; its cost is Monte Carlo, and a thin frontier event (an entrant that barely beats its pool) leaves few swaps to count. Table 1 reports it beside the other two: it certifies a dominated entrant 0.5 to 2.6% of the time with a near-copy rival and 1.0 to 3.4% with independent ones, and a truly better entrant 11 to 77% of the time, on 3000 boards per row.

Proposition 7 (no pairwise certificate holds its level)**.** *Let a certificate of Pareto membership be any test that depends on the data only through the entrant* b *and the observed most accurate cheaper system* j^**, and that rejects with probability tending to one when* b *beats* j^* *on every task on which they differ and the number of such tasks grows without bound. Then for every such test there are configurations in which* b *is truly dominated by a cheaper system and the test's conditional rejection probability, given the frontier event, tends to one.*

Proof. Put in $C(b)$ a system k with $\mu_k = \mu_b$, independent of b, and a copy j of b whose passes are flipped to failures independently with probability δ_T, with $\delta_T\sqrt{T} \to 0$ and $\delta_T T \to \infty$ (for instance $\delta_T = T^{-3/4}$). Then k dominates b. Given the frontier event, $j^* = j$ with probability tending to one, because $\bar{y}_b - \bar{y}_j$ is of order δ_T while $\bar{y}_b - \bar{y}_k$, conditional on being positive, is of order $T^{-1/2}$.

Against j, b wins every discordant task and the number of such tasks is of order $\delta_T T \to \infty$, so the test rejects with probability tending to one. □

The construction is not exotic: a leaderboard that lists an agent twice, with a stronger and a weaker model or with two reasoning-effort settings, holds exactly such a pair, and HAL does. Any certificate of Pareto membership must therefore look at the whole cheaper pool, which is what Propositions 5 and 6 do.

Corollary 1 (a chain of entrants)**.** *Let the step index t count the dates on which at least one entrant arrived, and test every entrant of the t-th such date at level α_t (alpha-spending) or at α_t plus the previous such date's level when every entrant of that date was certified (fallback), with its p-value multiplied by its per-date factor g. Then the family-wise error of the chain of certificates is at most α, in the sense of Proposition 4 when the pairwise p-values are used and in the sense of Proposition 5 when the composite ones are.*

Proof. A date's level is fixed by the earlier dates alone, before any outcome of the date is seen. On a date with g submissions and $m \le g$ entrants, the probability that some null entrant is certified is at most $\sum_{\text{null entrants}} \alpha_t / g \le \alpha_t$, by Proposition 4 and the union bound; summing over dates gives at most $\sum_t \alpha_t = \alpha$. For the fallback, consider the event that date t is the first date on which a null entrant is certified. Every earlier date whose level was passed along the run that feeds t had all its entrants certified, and none of them null, so the run consists of non-null dates; the runs that feed different first-failure dates are disjoint, and the levels available to null dates sum to at most α, as in Corollary 2 of Banahene and Opoku [5]. Passing a date's level on when only some of its entrants were certified would break this: a date holding a certified non-null entrant and an uncertified null one would sit inside the run and have its level counted twice. □

Indexing each same-date entrant by its own step would make an entrant's level depend on its rank within the date, which is settled only after the date's results are seen; counting dates avoids this and spends one level per date.

Remark 4 (random costs)**.** *When costs are measured per task, $C(b)$ is itself random, and b's mean cost comes from the same run as its outcomes (tokens are spent on failed tasks too), so the holder is not strictly free of b. For certification we fix each system's cost at its mean on the day of its arrival and check the consequence by resampling every system's per-task costs and rebuilding the chain: on the boards with per-task costs, an entrant keeps the same holder in 0.82 to 1.00 of 200 resamples, the number of entrants moves within 50 to 73 around 59, and the number certified under alpha-spending within 5 to 7 around 5 (Section 6). The frontier set of Section 3.3 is where the cost uncertainty is carried in full.*

3.8 The frontier winner's curse

A system reported as "on the frontier" was selected for beating every cheaper system, and its reported accuracy is inflated by the selection, like a leaderboard winner's. The selection event, $\bar{y}_b \ge \bar{y}_j$ for all $j \in C(b)$, is a polyhedron in the vector of accuracies, so the conditional method and the iterated bootstrap of Banahene and Opoku [5] apply with the pool $C(b) \cup \{b\}$ in place of the whole board. The conditional estimate is the median-unbiased estimate under the truncated normal law of $\bar{y}_b$ given that it is the largest of its pool [1, 15]. The bootstrap finds the winner of the pool again in every task resample and takes its mean excess over its own observed accuracy as the bias; a second bootstrap level removes the bias the first correction leaves. Costs are fixed at

				Bias, points (standard error)				Coverage of the 95% interval			
Systems	Tasks	Winners	Pool	reported	bootstrap	iterated	conditional	reported	bootstrap	iterated	conditional
8	50	796	4.6	+1.3 (0.2)	-0.7 (0.2)	-0.8 (0.3)	-0.7 (0.3)	0.95	0.91	0.90	0.96
8	150	841	4.6	+0.8 (0.1)	+0.0 (0.1)	+0.0 (0.1)	-0.2 (0.2)	0.94	0.92	0.90	0.95
8	500	836	4.8	+0.3 (0.1)	+0.0 (0.1)	+0.0 (0.1)	-0.2 (0.1)	0.95	0.94	0.93	0.96
20	50	1242	9.3	+2.2 (0.2)	-0.5 (0.2)	-0.7 (0.2)	-0.4 (0.2)	0.93	0.90	0.89	0.96
20	150	1319	9.2	+1.1 (0.1)	+0.2 (0.1)	+0.2 (0.1)	-0.1 (0.1)	0.94	0.92	0.91	0.95
20	500	1328	9.1	+0.3 (0.1)	-0.0 (0.1)	-0.0 (0.1)	-0.3 (0.1)	0.95	0.93	0.92	0.95

Table 2: The frontier winner's curse on known truth, 300 boards per scenario, every frontier member with at least one cheaper rival. Winners: the number of such members over the 300 boards; pool: mean number of systems that cost no more. Bias: mean of estimate minus true accuracy, in points, with its Monte Carlo standard error. Coverage: share of 95% intervals that contain the true accuracy.

their means, as in certification. A system with no cheaper rival, the cheapest on the board, has a pool of one and nothing to correct.

Table 2 checks both corrections on the simulated boards of Section 5. The reported accuracy of a frontier member overshoots the truth by +0.3 to +2.2 points on average, the larger values on the crowded boards of 20 systems; the iterated bootstrap leaves a bias of at most 0.8 points and the conditional estimate at most 0.7. The reported interval's coverage falls to 0.93; the conditional interval covers 0.95 to 0.96, while the bootstrap interval covers only 0.89 to 0.93, below nominal, as the companion paper found for the leaderboard winner. The frontier curse is milder than the leaderboard winner's curse of the companion paper because the pool is smaller than the board, and it is strongest for the expensive systems at the top of the staircase, whose pools are largest. Only 75 to 92% of the frontier members with a rival are truly undominated, which is the plain frontier's error rate seen from the other side.

3.9 What is proved and what is simulated

Table 3 says, for each tool, what is proved under which conditions and what rests on simulation or held-out data.

4 Data: HAL with per-task costs rebuilt

HAL publishes, for every run it hosts, an encrypted trace archive in its Hugging Face dataset (`agent-evals/hal_traces`), about 113 GB across 380 archives at the time of the audit. Each archive holds the run's configuration, its per-task outcomes, its total cost as HAL computed it, and the full call log: every model call the agent made, tagged with the task it served and the tokens it used. The decryption key is the public one in HAL's own harness. We read only the results member of each archive, a few hundred kilobytes to a few hundred megabytes, by HTTP range requests, and reduced each one on the fly to a compact record of outcomes, token counts and costs per task. 5 archives whose results member was too large to decrypt in memory on our machine (the largest 12.8 GB) were skipped; the other 375 were read.

Per-task cost. A task's cost is the priced token usage of the model calls logged under it, with HAL's own formula (fresh input tokens at the model's input price, cache reads and writes at the model's cache price, output tokens at the output price), one price table, HAL's current one, for every run, and each model call counted once. The last clause matters. HAL's reported total sums

Tool	Proved	Checked by simulation or held-out data
Frontier band	Covers the whole frontier curve with probability $\geq 1-\alpha$, asymptotically, for fixed or per-task costs and with no condition on ties (Prop. 1)	Coverage of the true curve on known truth with per-task costs and closely spaced true costs (Table 4)
Frontier set	Contains the whole true frontier with probability $\geq 1-\alpha$, asymptotically, fixed costs (Prop. 2); per-task costs without ties (Prop. 3)	Retention with per-task costs and against six alternative sets (Tables 4 and 5); held-out halves of the real boards (Table 6); run-to-run noise and task clusters (Section 7)
Undominated frequency	Nothing; a description	Its mean over truly undominated and truly dominated systems
Cheapest plausible best	Coverage of the best set, from Hsu [12]	Nothing further
Certifying an entrant	Finite-sample selective validity of the record claim against the observed best cheaper system (Prop. 4); a Pareto certificate under the normal approximation (Prop. 5) and an exact one (Prop. 6); no pairwise certificate holds its level (Prop. 7); chain-wise error (Cor. 1); costs fixed in advance	Level and power of both certificates with near-copy and independent rivals (Table 1); sensitivity to cost noise by resampling
Exact frontier sets	Per-system retention $\geq 1-\alpha$ for any T; the joint version contains the true frontier with probability $\geq 1-\alpha$ for any T (Section 3.3)	Their sizes against the bootstrap sets on known truth and on HAL
Frontier winner's curse	The conditional estimate is exact under normality for a fixed pool [15]	Bias and interval coverage on known truth for both corrections (Table 2)

Table 3: What each tool guarantees.

the usage block of every span in the trace, and an agent that calls a provider through a wrapper library logs two spans for one call, the wrapper and the provider call inside it, both carrying the same usage block. The effect was reported on the dataset's discussion board in September 2026 and confirmed by the HAL team [16]; HAL's pages now carry a caveat, but the totals stand. Of the 332 runs we read with usage, 141 have a reported token total of exactly twice the once-counted total, 44 fall between once and twice, 17 above twice, and 130 are unaffected; over all runs the reported totals overstate token use by 49%. Because the doubling depends on the agent's software stack and not on the model, it changes the cost ordering between agents on the same board, so a frontier drawn from HAL's own totals is wrong on every board that mixes the two kinds of run. We drop the duplicate spans (a call whose parent carries an identical usage block, or whose children's usage sums to its own), and the supplement lists the ratio for every run. As a check, our per-task costs sum to HAL's reported total to the cent on runs without nested spans that were made after the last price change of the models they used, and to exactly half of it on runs where every provider call was wrapped. Where the two differ otherwise it is because HAL's total embeds the prices in force when the run was made: an o3 run from May 2025 reports several times what the current

table gives for the same tokens, o3's price having fallen by four fifths in June 2025, and an early CORE-Bench run priced cached tokens at the full input rate. A frontier compares systems with one another, so it needs one price list and one way of counting; we reprice and recount everything and report HAL's figure beside ours in the released tables. The two effects compound: over the 318 runs costing at least a dollar, HAL's reported totals sum to $44,679 against $26,782 once counted and repriced, the median run is overstated by a factor of 2.00, 179 runs by more than half, and the largest gap is a factor of 31.

Outcomes and task sets. Outcomes come from the run's lists of successful and failed tasks or from its per-task evaluation results. A benchmark's task set is the tasks with an outcome in at least half of its runs; a run's task without a logged outcome counts as a fail, which touches 9 of the 356 kept runs, each missing at most 5% of its tasks. Reruns of one configuration and HAL's test uploads of the browser agents are separate systems, as HAL lists them; they are each other's rivals at near-identical cost and add to the dominated-on-arrival count of Section 6. On ColBench backend programming HAL's accuracy is the share of tasks scored exactly one, and we use that; on ColBench frontend design HAL's metric is a continuous similarity score in $[0, 1]$, and we keep it, so "accuracy" there is a mean score.

Exclusions. Of the 375 run records, 356 enter the audit. 11 runs were dropped because they carried outcomes for fewer than 95% of the benchmark's tasks, and 8 because their call logs carried no model call on more than half of their tasks (for instance runs of Claude Code, which does not log through HAL's tracer), so that their cost is unknown rather than zero; a system with an unknown cost cannot be placed on a cost-accuracy frontier, and dropping it is the only honest option. A kept run's tasks that have an outcome but no logged model call, a logging gap or a run that failed before its first call, are priced at the run's mean cost over its costed tasks rather than at zero, since a zero would push the run toward the frontier: 47 kept runs have at least one such task, 7 have more than a tenth, and 1.5% of all system-task pairs are imputed; the supplement gives the count per system. On Online-Mind2Web the browser agents' logs carry a call record per step but no token usage on any call, so no per-task cost exists for any run there and there are no per-call usage blocks to double count; the run totals, which HAL computed from the stored run-level usage, reprice exactly under the current table (except for two o3 runs, for the reason above), and we use each run's repriced total spread evenly over its tasks. On that board cost is therefore a fixed number per system, not resampled, and it is marked with a dagger in every table and figure.

Why only HAL. Terminal-Bench 2 publishes cost for three of 244 job files, and its current public board (version 4.0) gives cost per submission but no per-task results; SWE-bench Verified's cost metadata covers 47 submissions that have no per-task results in the experiments repository. Neither supports paired inference, so the audit rests on HAL's 11 benchmarks, which is also the cleanest case for the method, with per-task outcomes and per-task costs for every kept run on 10 of them.

5 Validation on known truth

Boards were simulated with $K \in \{8, 20\}$ systems and $T \in \{50, 150, 500\}$ tasks, true accuracies and mean costs spread so that several systems are truly undominated, task difficulty shared across systems so that outcomes are correlated within a task, and per-task costs that vary around each system's mean. On each of 300 boards per scenario the observed frontier and the frontier set were

				Frontier set			Band	Plain frontier		Undominated freq.	
Systems	Tasks	Costs	Corr.	kept, each	kept, all	size (true)	covers	falsely on	missed	true	dominated
8	50	per task	0.6	0.994	0.98	7.1 (4.0)	0.96	12%	17%	0.75	0.15
8	150	per task	0.6	0.990	0.96	6.2 (4.0)	0.98	7%	10%	0.85	0.10
8	500	per task	0.6	0.994	0.98	5.3 (3.9)	0.98	5%	4%	0.92	0.08
20	50	per task	0.6	0.997	0.98	16.9 (5.8)	0.97	21%	29%	0.62	0.10
20	150	per task	0.6	0.995	0.98	13.3 (5.8)	0.98	13%	16%	0.76	0.07
20	500	per task	0.6	0.987	0.93	10.1 (5.9)	0.98	8%	9%	0.86	0.05
20	150	fixed	0.6	1.000	1.00	13.4 (5.9)	0.98	10%	13%	0.80	0.06
20	150	per task	0.3	0.989	0.95	10.7 (4.4)	0.99	12%	13%	0.80	0.05
60	45	per task	0.6	0.997	0.98	48.5 (7.9)	0.97	28%	39%	0.51	0.05

Table 4: Known-truth simulation, 300 boards per scenario. Kept, all: share of boards on which every truly undominated system is kept in the frontier set (the guarantee of Proposition 2; target at least 0.95). Kept, each: share of truly undominated systems kept, one by one. Size: mean size of the set, with the true frontier's size in brackets. Band covers: share of boards on which the band of Section 3.2 contains the true frontier curve at every evaluated cost. Plain frontier: share of its members that are truly dominated, and share of the true frontier it misses. Undominated frequency: mean over truly undominated and truly dominated systems.

computed from the sample and compared with the true frontier. Table 4 has the results. Every truly undominated system was kept at once on 0.93 to 1.00 of boards: at or above the nominal 0.95 in eight of the nine scenarios and 0.93 in the ninth (20 systems on 500 tasks), where a rerun with 2000 boards gives 0.953; the standard error of a share near 0.95 is 0.013 with 300 boards. A given truly undominated system was kept 0.987 to 1.000 of the time. The plain frontier, over the same boards, wrongly included 5 to 28% of its members and missed 4 to 39% of the true frontier. The price of the guarantee is size: with 20 systems the set held 16.9 systems on average against a true frontier of 5.8 at $T = 50$, 13.3 against 5.8 at $T = 150$, and 10.1 against 5.9 at $T = 500$. The set does not become informative until a board has a few hundred tasks, which is the central practical message of the paper. Three further scenarios probe the assumptions: costs fixed at their means (the case Proposition 2 covers), a weaker link between cost and ability, and a crowded board of 60 systems on 45 tasks, the shape of CORE-Bench hard. Retention stays at 0.989 to 1.000; with fixed costs the set holds 13.4 systems against 13.3 with per-task costs on the same board shape, and on the crowded board it holds 48.5 of 60 systems against a true frontier of 7.9, with 28% of the plain frontier truly dominated, which is the picture the audit finds on CORE-Bench hard.

Table 5 puts the set beside the alternatives on four of the nine scenarios; the ranges below span all nine. The joint margin keeps the whole true frontier 0.93 to 1.00 of the time; the exact joint set does too, at a larger size; the two per-system variants fall to 0.87 on the crowded board. The two shortcuts fail where the boards are small or crowded, which is where a leaderboard needs the set most: unadjusted pairwise tests keep the whole true frontier 0.78 to 0.98 of the time, and the frequency rule 0.55 to 0.99, because neither accounts for the number of rivals a system faces; on the large, sparse boards (20 systems on 500 tasks) the frequency rule does as well as the set with fewer systems, since there the plain frontier is nearly right already. With 20 systems on 50 tasks the sets hold 16.9 (joint), 18.2 (exact joint), 12.6 (unadjusted) and 10.4 (frequency) systems against a true frontier of 5.8.

5.1 Held-out halves of the real boards

Simulation has a known truth but a made-up world. Table 6 repeats the two checks on the real boards with at least 100 tasks, with the tasks split at random into two halves 200 times: the frontier

Set	$K = 20$, $T = 50$ kept, all	size	$K = 20$, $T = 150$ kept, all	size	$K = 20$, $T = 500$ kept, all	size	$K = 60$, $T = 45$ kept, all	size
Confidence set (joint margin)	0.98	16.9	0.98	13.3	0.93	10.1	0.98	48.5
Largest per-system margin	0.95	14.9	0.96	11.8	0.91	9.1	0.94	39.9
Per-system margins	0.91	14.2	0.92	11.1	0.89	8.7	0.87	35.8
Exact, per system	0.95	15.8	0.94	11.8	0.90	9.0	0.93	43.5
Exact, joint	0.99	18.2	0.98	14.0	0.94	10.4	1.00	55.3
Unadjusted pairwise tests	0.89	12.6	0.90	9.7	0.88	7.8	0.78	26.7
Frequency $\geq \alpha$	0.81	10.4	0.94	9.1	0.96	7.9	0.55	16.9
True frontier		5.8		5.8		5.9		7.9

Table 5: Seven ways to turn the same board into a set, on known truth (300 boards per scenario, per-task costs). Kept, all: share of boards on which every truly undominated system is kept (the joint guarantee; target 0.95). Size: mean number of systems kept. The confidence set of Section 3.3 is the first row; the next two are its per-system variants; the exact rows use one-sided paired tests with Bonferroni over each system's rivals or over every rival pair; "unadjusted" keeps a system unless some cheaper rival beats it at the 5% level with no correction; "frequency" keeps a system that is on the observed frontier in at least 5% of the task resamples.

			B's frontier members		Bias, points			Coverage		
Benchmark	Tasks	Winners	in A's set	on A's frontier	reported	iterated	conditional	reported	iterated	conditional
ColBench backend	1000	1469	1.00	0.82	+0.6	-0.0	-0.1	0.95	0.92	0.96
ColBench frontend	100	706	1.00	0.68	+0.7	+0.4	+0.1	0.87	0.88	0.91
GAIA	165	1582	1.00	0.61	+2.3	+0.1	+0.2	0.89	0.83	0.92
Online-Mind2Web†	300	1011	1.00	0.71	+1.6	-0.2	-0.2	0.91	0.89	0.96
ScienceAgentBench	102	684	1.00	0.64	+3.4	+0.8	+1.3	0.92	0.88	0.96
USACO	307	608	1.00	0.99	+0.1	+0.0	+0.0	0.97	0.97	0.97

Table 6: Held-out check on the HAL boards with at least 100 tasks, 200 random half splits each. The frontier, the frontier set and the curse corrections are computed on half A; half B is fresh data. Left: of the systems on B's own observed frontier, the share kept in A's frontier set and the share on A's plain frontier. Right: for A's frontier members with a cheaper rival (winners), the bias of the A-half reported accuracy and of the two corrected estimates against the B-half accuracy, shifted by the mean over all systems of the B-half minus A-half accuracy so that the halves' difficulty difference cancels, and the coverage of each 95% interval.

set, the plain frontier and the curse corrections are computed on one half, and the other half is scored as if it were a fresh benchmark. Of the systems that sit on the fresh half's own frontier, A's frontier set keeps 1.00 to 1.00, while A's plain frontier contains only 0.61 to 0.99 of them. For the 6060 frontier winners across boards and splits, the reported accuracy overshoots the fresh half by +0.1 to +3.4 points on average; the iterated bootstrap is within 0.8 points of it and the conditional estimate within 1.3. The conditional interval covers the fresh-half accuracy at least 0.91 of the time and the bootstrap interval at least 0.83, below nominal on the boards of about a hundred tasks, where fifty-task halves are thin. The two halves of a benchmark share its task pool, so this is a check against task-sampling noise only, which is the noise the method models.

6 Audit of the Holistic Agent Leaderboard

HAL's own frontiers do not survive consistent costing. Before any uncertainty enters, the point-estimate frontier itself depends on how costs are counted and priced. Table 8 draws it three

Benchmark	Systems	Tasks	Observed frontier	Frontier set	Best set	Top system (%, $/task)	Cheapest plausible best (%, $/task)
assistantbench	32	33	3	32	32	15.2, 4.39	3.0, 0.02
colbench backend programming	22	1000	9	14	5	43.7, 0.20	38.6, 0.02
colbench frontend design	14	100	4	7	5	82.9, 0.12	81.6, 0.03
corebench hard	60	45	8	53	1	75.6, 6.08	75.6, 6.08
gaia	36	165	10	22	7	74.5, 1.13	64.2, 0.74
online mind2web†	32	300	6	23	16	42.3, 0.57	36.3, 0.30
scicode	37	65	3	37	37	9.2, 0.72	1.5, 0.00
scienceagentbench	25	102	4	21	17	33.3, 0.06	27.5, 0.02
swebench verified mini	39	50	6	26	11	72.0, 1.57	54.0, 0.67
taubench airline	47	50	7	40	26	66.0, 3.14	44.0, 0.04
usaco	12	307	4	5	1	69.7, 0.11	69.7, 0.11

Table 7: Frontier audit of HAL's benchmarks at $\alpha = 0.05$, $B = 4000$ task resamples. Observed frontier: systems no cheaper system beats on the point estimates. Frontier set: systems that cannot be excluded from the frontier. Best set: systems that cannot be excluded as the most accurate. Costs are dollars per task, repriced from the call logs with one price table. †: no per-task costs exist on this board; each run's repriced total is spread evenly over its tasks and treated as fixed.

	Plain frontier size under			Reported to ours		Changes due to	
Benchmark	reported	once-counted	ours	lost	gained	counting	pricing
assistantbench	3	3	3	1	1	0	2
colbench backend programming	9	9	9	0	0	0	0
colbench frontend design	3	3	4	0	1	0	1
corebench hard	5	5	8	2	5	0	7
gaia	5	5	10	1	6	0	7
online mind2web†	6	6	6	0	0	0	0
scicode	3	3	3	1	1	0	2
scienceagentbench	8	6	4	4	0	2	2
swebench verified mini	4	6	6	0	2	2	0
taubench airline	6	6	7	1	2	0	3
usaco	4	4	4	0	0	0	0
All	56	56	64	10	18	4	24

Table 8: The plain frontier of each board under three costings of the same runs: HAL's reported totals, those totals with the duplicate spans removed (HAL's historical prices kept), and the once-counted, repriced costs of the audit. Lost and gained: systems that leave or join the frontier going from HAL's costing to ours; the last two columns split the changes into those made by counting each call once and those made by repricing after that.

ways. Going from HAL's reported totals to once-counted, repriced costs changes the frontier on 8 of 11 boards: 10 systems leave it and 18 join, 28 changes against 56 systems on the reported frontiers (50%). Removing the duplicate spans alone moves 4 systems on 2 boards, because the doubling is shared by the agents that use the same software stack and mostly leaves their order intact; pricing every run under one table moves 24 more, because HAL's totals carry the prices of the month each run was made, and because 5 runs are reported at a total near zero although their call logs price at more than a cent per task, all of them sitting on HAL's frontier for that reason alone. A reader of HAL's pages is looking at frontiers that mix two counting rules and a year of price changes. Figure 3 shows the two boards that change most.

Table 7 and Figures 1 and 2 give the audit proper. Four readings.

The frontier is far less certain than the picture suggests. Across the 11 benchmarks 64 of 356 systems sit on the observed frontier, but 280 cannot be excluded from it. The simultaneous

Benchmark	Tasks	Systems kept						Tasks needed	
		Plain frontier	joint set	largest-margin	per-system	exact, per system	exact, joint	to exclude half	for a 5-point lead
assistantbench	33	3	32	32	32	32	32	171	442
colbench backend programming	1000	9	14	12	12	12	15	2,857	1,012
colbench frontend design	100	4	7	7	7	7	7	7,863	52
corebench hard	45	8	53	49	49	52	57	408	1,630
gaia	165	10	22	20	20	21	23	606	1,699
online mind2web†	300	6	23	21	19	20	24	1,811	1,370
scicode	65	3	37	37	37	37	37	265	365
scienceagentbench	102	4	21	18	18	20	21	529	1,169
swebench verified mini	50	6	26	23	17	24	27	122	1,179
taubench airline	50	7	40	35	32	39	42	291	1,748
usaco	307	4	5	5	5	5	6	1,057	697

Table 9: Five frontier sets and two resolution numbers per board. Joint set: the confidence set of Section 3.3; largest-margin and per-system: its two per-system variants; exact, per system and exact, joint: the finite-sample sets of Section 3.3. Tasks to exclude half: the task count at which half of the systems now in the set but off the plain frontier would leave it, if their gaps to their strongest cheaper rival stayed as observed. Tasks for a 5-point lead: the task count at which a cheaper system leading by five accuracy points would be excluded, given the board's median paired noise.

margin is 3.04 to 3.77 standard errors of a paired difference, and on a board of 50 tasks a paired standard error is several accuracy points, so a system is ruled off the frontier only when a cheaper one leads it by ten to fifteen points.

The band says the same thing as a curve. The shaded region in Figure 1 is the 95% band of Section 3.2. Its half-width at a frontier member is 2.95 to 3.61 of that member's own standard error, and its width, taken as the median over the costs at which it is evaluated, runs from 6 points on ColBench frontend to 67 points on CORE-Bench hard. A reader who wants one picture of what a board can and cannot say about the best accuracy available at each price has it here, and the picture is humbler than the staircase. Even the best-measured boards, ColBench backend with 1000 tasks and USACO and Online-Mind2Web with about 300, have bands of 10 to 18 points; ColBench frontend is narrow (6 points) only because its continuous scores vary little from task to task; and on the boards with 33 to 65 tasks the band is 21 to 67 points wide, a large part of the whole range of accuracies. A step of the staircase can be trusted only where it is taller than the band.

Task count decides what can be said. The two boards where the set nearly equals the observed frontier are USACO (307 tasks: frontier 4, set 5) and ColBench backend programming (1000 tasks: frontier 9, set 14). The two boards where nothing can be excluded are AssistantBench (33 tasks, accuracies up to 15.2%) and SciCode (65 tasks, up to 9.2%): with success rates that low and samples that small, no cheaper system can be shown to beat any other. CORE-Bench hard, with 60 systems on 45 tasks, keeps 53 of them in the set.

How far the boards are from resolving their frontiers. Table 9 turns the set sizes into task counts. Excluding half of the systems that now sit in the set but off the plain frontier would take 122 to 7,863 tasks across the boards that have such systems; excluding a cheaper system that leads by five points would take 52 to 1,748. Like for like, the exact sets are the larger ones: the exact joint set holds 291 systems against the confidence set's 280 (larger on 7 boards, smaller on 0), and the exact per-system set 269 against the per-system set's 248, which is the usual trade between a finite-sample guarantee and power. The two per-system variants of the confidence set hold 259 and 248 systems against its 280: the joint guarantee costs a few dozen systems across the audit. The two boards with the largest "tasks to exclude half" (ColBench backend and frontend) are the

Benchmark	On frontier	With a cheaper rival	Shrinkage, iterated bootstrap (pt) mean	Shrinkage, iterated bootstrap (pt) max	Shrinkage, conditional (pt) mean	Below strongest cheaper rival after adjustment
assistantbench	3	2	3.3	5.3	1.1	1
colbench backend programming	9	8	0.3	0.5	0.1	0
colbench frontend design	4	3	0.2	0.5	0.4	1
corebench hard	8	7	4.3	7.7	3.5	4
gaia	10	9	1.2	2.4	1.3	2
online mind2web†	6	5	0.6	0.9	0.1	0
scicode	3	2	0.2	0.5	0.1	0
scienceagentbench	4	3	0.8	1.6	0.1	0
swebench verified mini	6	5	1.0	2.3	0.1	0
taubench airline	7	6	4.4	9.4	2.8	3
usaco	4	3	0.0	0.1	0.0	0
All	64	53	1.7	9.4	1.1	11

Table 10: Selection-adjusted accuracy of the systems on HAL's observed frontiers. Shrinkage: reported accuracy minus the adjusted estimate, in points, over the frontier members that have a cheaper rival. The last column counts frontier members whose iterated-bootstrap estimate falls below the reported accuracy of their strongest cheaper rival. Every member is listed in the supplement.

ones where the few unexcluded systems are near-ties with their rivals, which no task count resolves quickly.

The best is often clear even when the frontier is not. The best set is a single system on 2 boards (CORE-Bench hard and USACO) while the frontier set on those boards is larger. This is by construction: a system's rivals for the best set are all the other systems, and for the frontier set only the cheaper ones, with nearly the same margin, so the frontier set contains the best set, and a cheap, weak system stays in it whenever no cheaper system beats it by the margin. The frontier set is a statement about every system's own neighbourhood of cheaper rivals, and on most boards those neighbourhoods are thinly measured.

The cheapest plausible best is usually not the top system. On 7 of 11 boards the cheapest system that cannot be told apart from the best is a different and cheaper system than the most accurate one; on 2 further boards (AssistantBench and SciCode) the best set is the whole board, so the cheapest plausible best is simply the cheapest system and the statement is empty. The gaps are large: on tau-bench airline the top system costs $3.14 per task and scores 66.0%, while the cheapest plausible best costs $0.04 and scores 44.0%; on GAIA the pair is $1.13 at 74.5% against $0.74 at 64.2%; on SWE-bench Verified mini, $1.57 at 72.0% against $0.67 at 54.0%. A reader should not take this as "the cheap system is as good"; it says that 50 tasks cannot show that it is worse, at a level that accounts for all 47 systems being compared at once. The right response is more tasks, not a cheaper purchase.

The frontier winner's curse on HAL. Table 10 applies the corrections to the 64 systems on HAL's observed frontiers, 53 of which have a cheaper rival. The iterated bootstrap takes 1.7 points off a frontier member's reported accuracy on average and 9.4 at most; the conditional estimate takes 1.1 on average. For 11 of the 64 members, on 5 boards, the adjusted accuracy falls below the reported accuracy of the strongest cheaper rival: at its adjusted accuracy the system would not be on the frontier at all. The corrections are largest where the tasks are few and the frontier members many, CORE-Bench hard and tau-bench airline, and negligible on USACO and ColBench backend, where the tasks are many.

Benchmark	Systems	Cheapest so far	Entrants	Dominated	Record claim (pairwise)			Pareto certificate (normal)			Pareto certificate (exact)		
					unadj.	spending	fallback	unadj.	spending	fallback	unadj.	spending	fallback
assistantbench	32	2	1	29	0	0	0	0	0	0	0	0	0
colbench backend programming	22	3	7	12	3	2	2	3	2	2	1	0	0
colbench frontend design	14	3	1	10	0	0	0	0	0	0	0	0	0
corebench hard	60	6	12	42	1	0	0	1	1	1	1	0	0
gaia	36	5	11	20	1	0	0	1	0	0	0	0	0
online mind2web†	32	6	10	16	1	1	1	1	1	1	1	1	1
scicode	37	4	0	33	0	0	0	0	0	0	0	0	0
scienceagentbench	25	3	5	17	0	0	0	0	0	0	0	0	0
swebench verified mini	39	5	10	24	2	1	1	2	1	1	2	1	1
taubench airline	47	5	7	35	1	0	0	1	0	0	1	0	0
usaco	12	1	5	6	3	2	2	4	3	3	2	2	2
All	356	43	69	244	12	6	6	13	8	8	8	4	4

Table 11: Frontier entrants on each HAL board, walking the submissions in date order with each system's mean cost per task fixed on arrival. Cheapest so far: the system was cheaper than everything before it and joined the frontier with no rival to beat (no test). Entrants: the system beat every cheaper system on arrival and was tested against the most accurate of them with the selective record test and the per-date factor. Dominated: a cheaper earlier system was at least as accurate, so the system was never on the frontier. Record claim: the pairwise selective test of Proposition 4; Pareto certificate: the composite test of Proposition 5. For each, certified at the 5% level with only the per-date factor applied (unadj.), under alpha-spending, and under the online fallback, the levels being spent per date with an entrant; spending and fallback control the family-wise error over the board's whole chain.

Certifying entrants on HAL's own timeline. Table 11 applies the rule of Section 3.7 to each board's history. Of 356 arrivals, 43 were the cheapest system so far and joined the frontier without a rival to beat, 244 were dominated on arrival by a cheaper system and were never on it, and 69 (19%) beat every cheaper system and were frontier entrants in the sense a leaderboard would announce. Tested against their most accurate cheaper rival, 12 of the 69 pass at the 5% level with only the per-date factor applied, and 6 under alpha-spending and 6 under the online fallback, the levels being spent over the 55 dates on which an entrant arrived. The composite test of Proposition 5, which certifies Pareto membership rather than a record claim, passes 13 at the 5% level and certifies 8 under alpha-spending and 8 under the fallback; 6 entrants are certified by both rules, 0 by the record claim only and 2 by the Pareto certificate only. The exact randomization form of the certificate (Proposition 6) passes 8 at the 5% level and certifies 4 under alpha-spending and 4 under the fallback. The certified steps, under either certificate, are the large, well-measured ones, 6 to 44 points on boards with at least 45 tasks. The rest are of three kinds. 20 entrants beat their rival by under five points on boards of at most 165 tasks, where such a step is within about one standard error. 47 of the 69 entrants arrived on a date with other submissions, so that the best-of-date selection is priced by the per-date factor. And 6 entrants pass the 5% level after the per-date factor but not the level left in the alpha-spending budget by the time they arrive, which is the price of controlling the error over the whole chain, as the companion paper found on its boards. A maintainer who announced every entrant as "new on the frontier" would have made 69 claims, of which 6 to 8 survive a guarantee, depending on the claim and the rule.

7 Limitations

Run-to-run noise. Each system has one run, and the standard errors carry task-sampling noise only. HAL does hold 8 pairs of runs of one configuration (same agent, model and settings) on 3 boards, which measure the missing piece directly: the two runs of a pair differ by 6.4 accuracy points on average and 14.0 at most, agree on 72 to 92% of their tasks, and differ in cost by a median

factor of 1.09 (at most 1.72). Pooled over the pairs, one run's run-to-run standard deviation of accuracy is 5.9 points, against a task-sampling standard error of 6.9 points for a single run on those boards: the two sources of noise are of the same size, and a leaderboard that runs each system once reports about 76% of the standard error its numbers carry. Carried to every board through the per-task disagreement rate (so that it shrinks with the task count as the sampling error does) and added to both systems of every paired comparison, run-to-run noise widens the margins by a factor of 1.23 to 3.70 and grows the frontier sets from 280 to 304 systems in all. Everything in the audit is therefore a lower bound on the uncertainty; the remedy is repeated runs, which the method accommodates by averaging them.

Task clusters. Tasks are treated as exchangeable draws. Where tasks cluster (several from one repository, one website, one problem family) the standard errors are optimistic, as in any task-level analysis. The module can resample task groups instead of tasks, but HAL's boards offer little to resample: SWE-bench Verified mini's 50 tasks come from 2 repositories and USACO's 307 from 4 difficulty divisions, and resampling units that few gives margins of 3.36 and 5.52 standard errors against 3.66 and 3.04, numbers that say more about having two or four units than about dependence. A benchmark with many small groups is the case where a cluster bootstrap would be informative, and none of these is one.

The rest. Costs depend on prices, which change; we fix one table and say which, and a reader pricing a run under another table should rebuild the frontier rather than reuse ours. Systems may have been tuned on the public tasks, which inflates every accuracy on the board and is outside the model. Outcomes are taken as HAL recorded them; HAL has since disclosed that on tau-bench airline its Generalist agent's episodes that never terminated were recorded as failures, so those rows are lower bounds on that scaffold's accuracy [16]. The guarantees of the band and the set are asymptotic. The band needs no condition on the costs; the set is proved for fixed costs and for per-task costs whose true means are distinct, and systems tied in true mean cost are the case left to simulation. On Online-Mind2Web the cost is a run total treated as fixed. The set uses one margin for the whole board, which is conservative when the systems are many; per-system margins would give a smaller valid set. Finally, the set is a statement about K specific systems on T specific tasks, and a new system or a new task changes it.

8 Recommendations

1. Report the frontier set beside the frontier. A board that draws a frontier from point estimates should say which systems cannot be excluded from it, at a stated level, how many tasks that statement rests on, and how many it would take to resolve it (Table 9: hundreds to thousands on the present boards).

2. Publish per-task costs. The confidence set needs them to carry the cost uncertainty, and the audit shows that run totals alone can hide a fourfold difference in price between two dates.

3. Price every run under one table and name the table and its date.

4. Give buyers the cheapest plausible best, with its level, rather than the top system.

5. Certify frontier entrants with an error budget spent over the calendar, and say which claim is certified: the record claim against the most accurate cheaper system (exact), or Pareto membership (the composite test, under the normal approximation), with costs fixed on arrival.

6. Count each model call once and price every run under one table before drawing a frontier; HAL's published frontiers change on most benchmarks when this is done.

9 Conclusion

A cost-accuracy frontier is an estimate. Treated as one, with a confidence set in place of a line, HAL's benchmarks say less than the lines suggest and more than nothing: on some boards the frontier is sharp, on most it is wide, and on a few it is empty of content until more tasks are run. The tools here let a maintainer say which, and let a new system be placed on the frontier with a guarantee that survives its own selection.

Reproducibility

Code, the compact per-task records derived from HAL's public traces, the simulation and the audit are in the paper's repository; the methods are in the open-source package `honestboard` (module `frontier`). Raw HAL archives are not redistributed.

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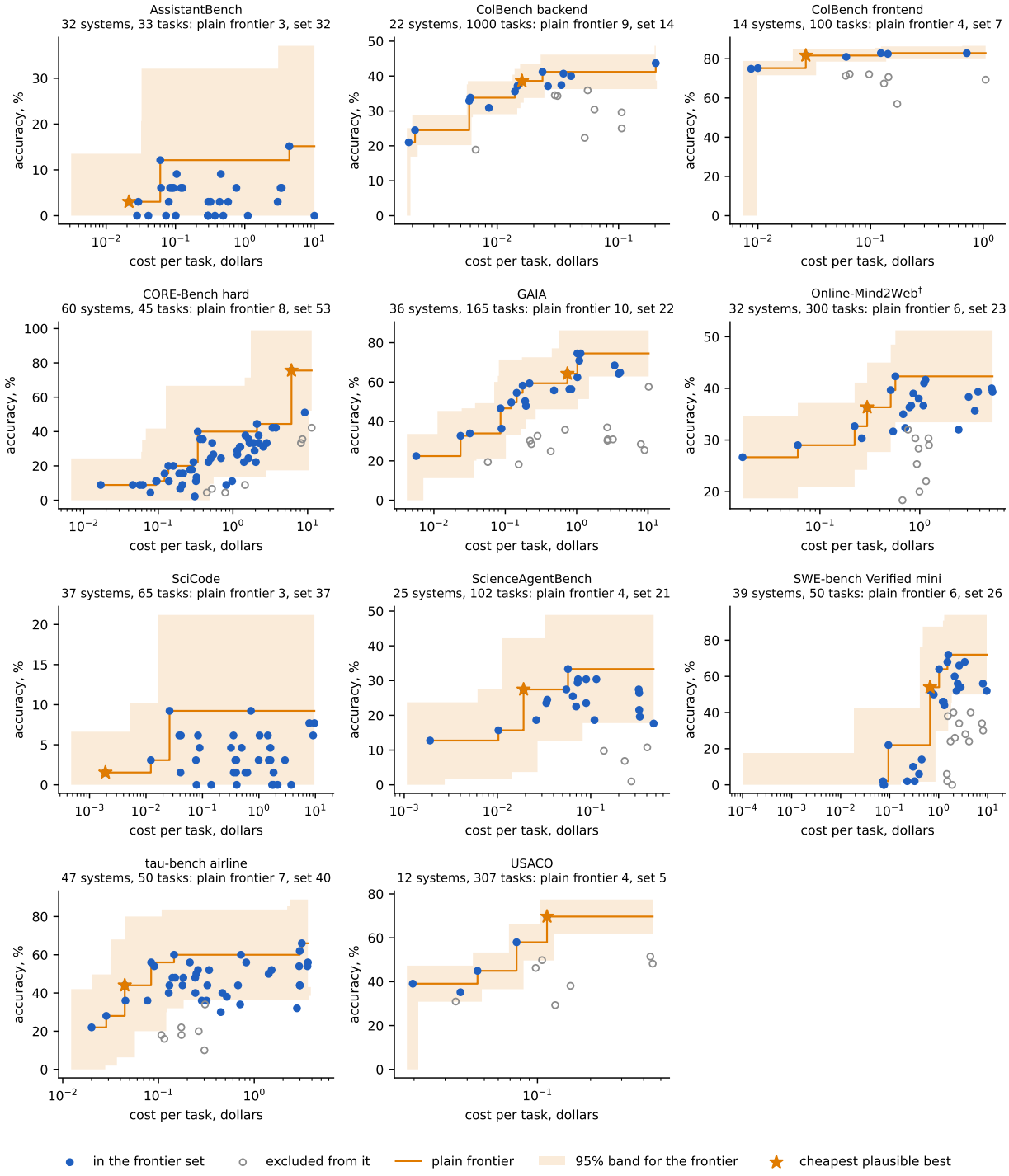


Figure 1: Cost per task against accuracy on each HAL benchmark. The orange staircase is the observed frontier. Filled points are systems that cannot be excluded from the frontier at the 5% level; hollow points can be. The star is the cheapest plausible best: the cheapest system that cannot be excluded as the most accurate. The shaded region is the 95% simultaneous band for the frontier curve (Section 3.2). Costs of zero are drawn at 10^{-4} dollars. The dagger marks the board whose costs are run totals spread over the tasks and treated as fixed.

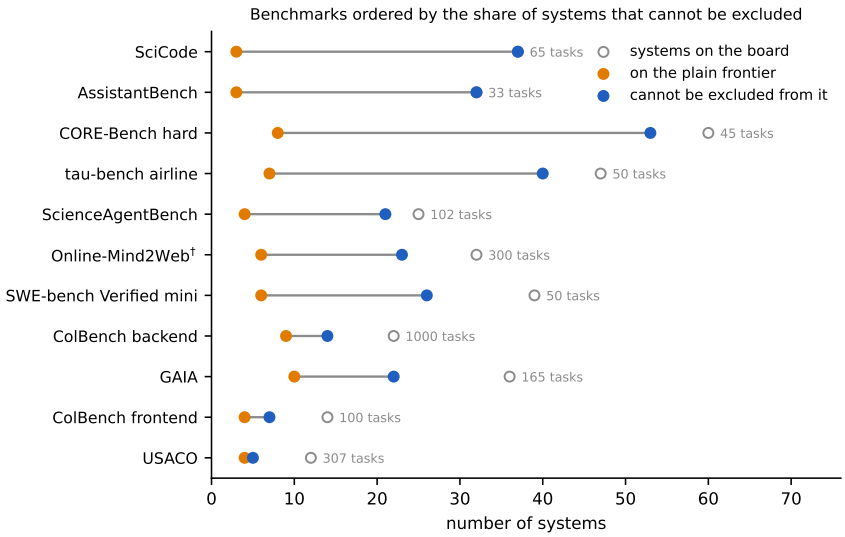


Figure 2: For each benchmark, the number of systems on the plain frontier (orange), the number that cannot be excluded from it at the 5% level (blue), and the number of systems on the board (hollow), with the task count beside it. Benchmarks are ordered by the share of their systems that cannot be excluded.

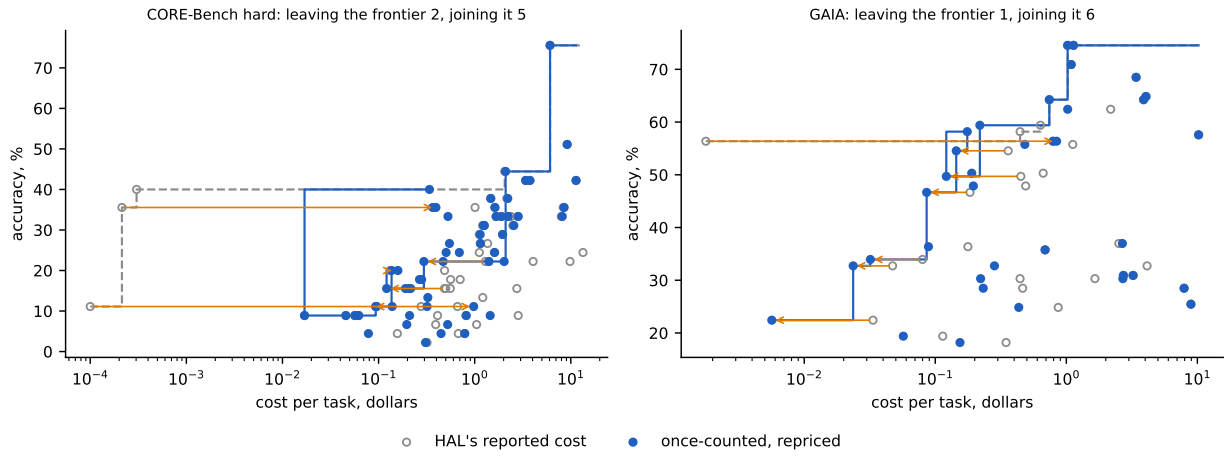


Figure 3: HAL's published frontier (hollow points, dashed staircase, HAL's reported costs) against the frontier under once-counted, repriced costs (filled points, solid staircase) on the two boards that change most. Arrows join the two costs of each system whose frontier status changes. Costs of zero are drawn at 10^{-4} dollars.